Ron Bibb
Independent Researcher
Lilburn, Georgia, USA
ronbibb@gmail.com

9 August 2026

Editors
General Relativity and Gravitation

Dear Editors,

Please consider my manuscript, "Surface-Energy Sign Classification for Timelike Kantowski–Sachs–Schwarzschild Junctions on the Ordinary Branch," for publication in *General Relativity and Gravitation*. The manuscript identifies an invariant causal-sector split. Under the ordinary orientation, every compatible finite-rapidity junction from homogeneous Kantowski–Sachs geometry to the retained Schwarzschild $F > 0$ exterior requires negative Israel surface density, and that ordering persists for a timelike crossing at $F = 0$. In the complementary $F < 0$ sector, with $\epsilon_P = +1$ and the increasing-R Schwarzschild side retained, the ordering becomes sign-indefinite: two distinct future-infalling local embeddings through the same bulk event realize opposite density signs, and an exact compatible choice has $p_s = \sigma/2$, strictly satisfying NEC, WEC, DEC, and the intrinsic shell SEC.

The contribution is the combined causal-sector classification, exact positive-density interval, and explicit regular-chart witness construction, rather than the underlying general BKT junction formula. The Schwarzschild horizon is the invariant boundary between the classified open sectors, while sign indeterminacy begins strictly at $F < 0$. The interior result is local and does not claim a global gluing, dynamics, or validation of generally spacelike transition layers. This combination of exact junction analysis, branch control, and energy-condition analysis is well aligned with the scope of *General Relativity and Gravitation*.

The manuscript is original, is not under consideration elsewhere, and has been prepared independently without institutional or grant support. The archived manuscript source, figure-generation scripts, and verification suite are available through the stable Zenodo archive at doi:10.5281/zenodo.21872425; development history remains public at github.com/RonBibb/junction-ec-paper.

Subject to the journal's conflict-of-interest checks, potential referees with relevant expertise include João L. Rosa, Sebastian Carloni, S. H. Mazharimousavi, José M. M. Senovilla, and S. Khakshournia.

Sincerely,

Ron Bibb
Independent Researcher
Lilburn, Georgia, USA
ronbibb@gmail.com
ORCID: 0009-0004-1153-2464